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A diagram showing three overlapping ellipses. The central ellipse is filled with a solid blue color. The two ellipses on either side are white with a black outline, representing a three-lobed structure.

Overview

Features

Feature	Description
Real-time Navigation	Dynamic maps that guide attendees through the event, avoiding congestion and highlighting points of interest.
Safety Alerts	Instant notifications about potential hazards, emergency exits, and evacuation routes.
Personalized Recommendations	Suggestions for events, activities, and amenities based on user preferences and location.
Reporting	Ability to report incidents or safety concerns with location data

Feature	Description
Crowd Density Monitoring	Real-time visualization of crowd density hotspots, enabling proactive management of potential bottlenecks.
Incident Management	Tools to log and resolve incidents, coordinate security personnel, and communicate with attendees.
Resource Allocation	Data-driven insights to optimize staffing, security, and amenities based on crowd movement and needs.
Communication Tools	Send targeted notifications to specific user groups about event updates, schedule changes, or potential hazards.

Administrators

Feature	Description
User Management	Tools to manage user accounts, roles, and permissions.
Event Configuration	Configure event details, maps, points of interest, and safety protocols.
Analytics Dashboard	Comprehensive overview of event performance, user engagement, and incident resolution.
System Monitoring	Tools to monitor the health and performance of the CrowdSafe platform.

Technology Stack

Frontend

- React
- TypeScript
- Vite
- Mapbox GL JS

Backend

- Node.js
- Express
- Firebase Cloud Functions

Database & Auth

- Firebase Firestore
- Firebase Authentication